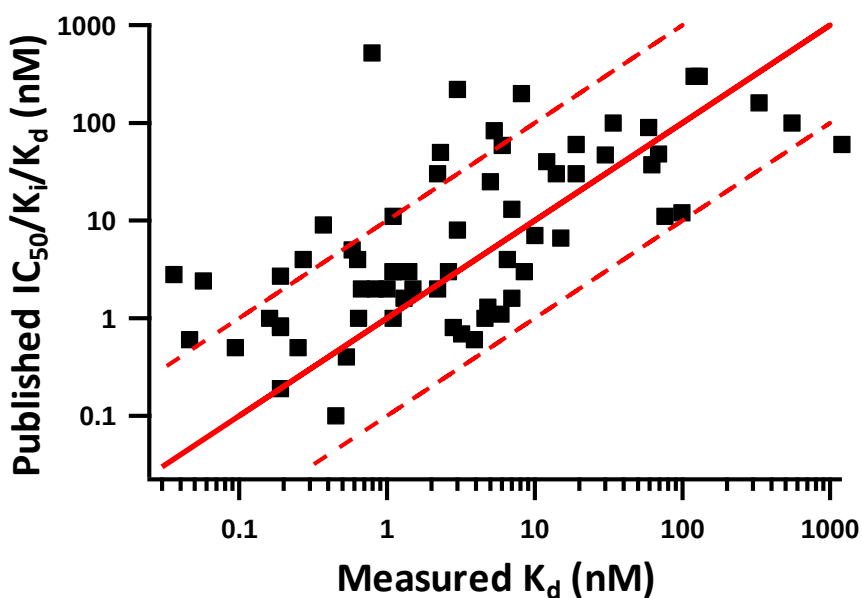
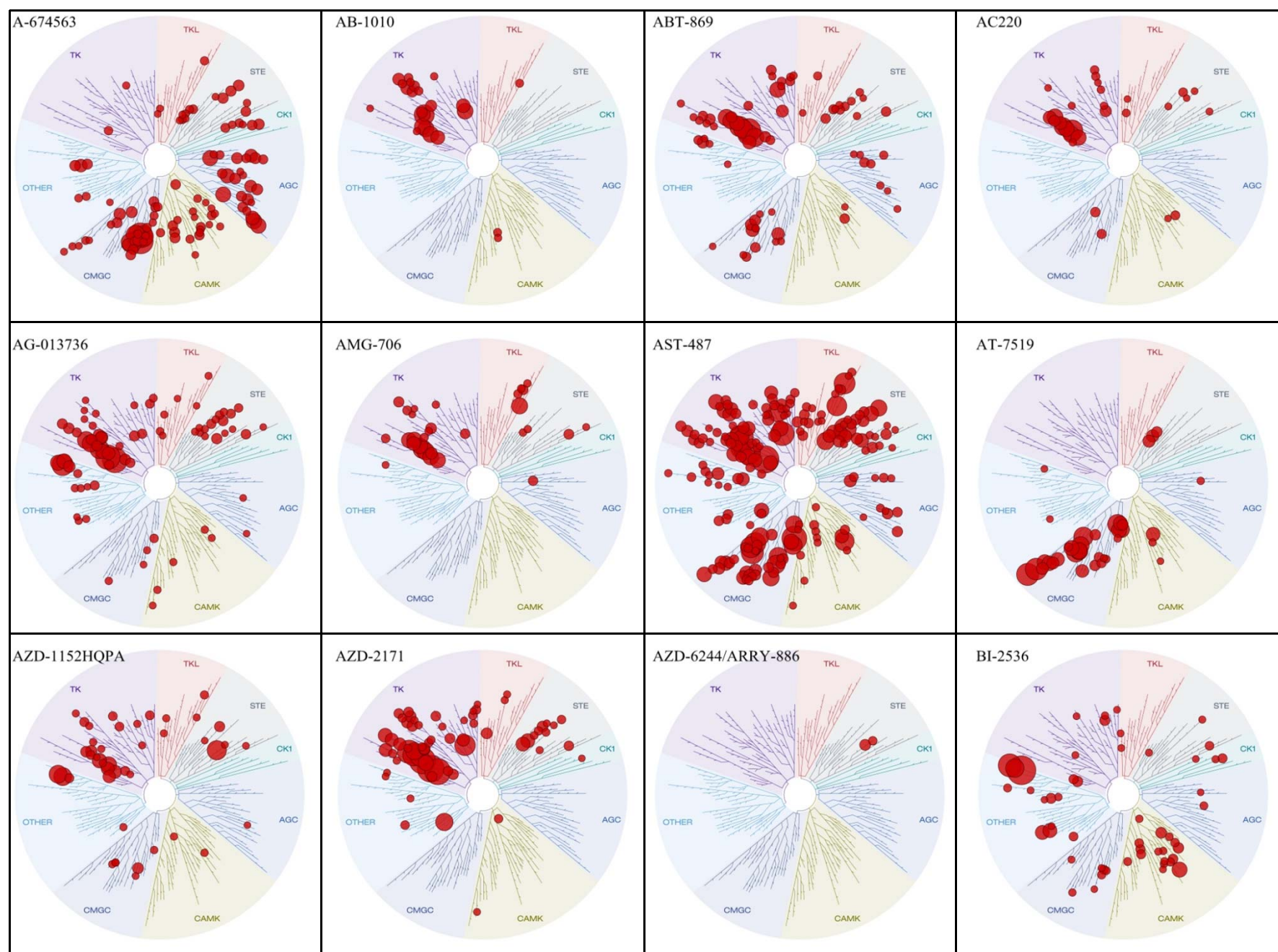


Supplementary Figure 1.

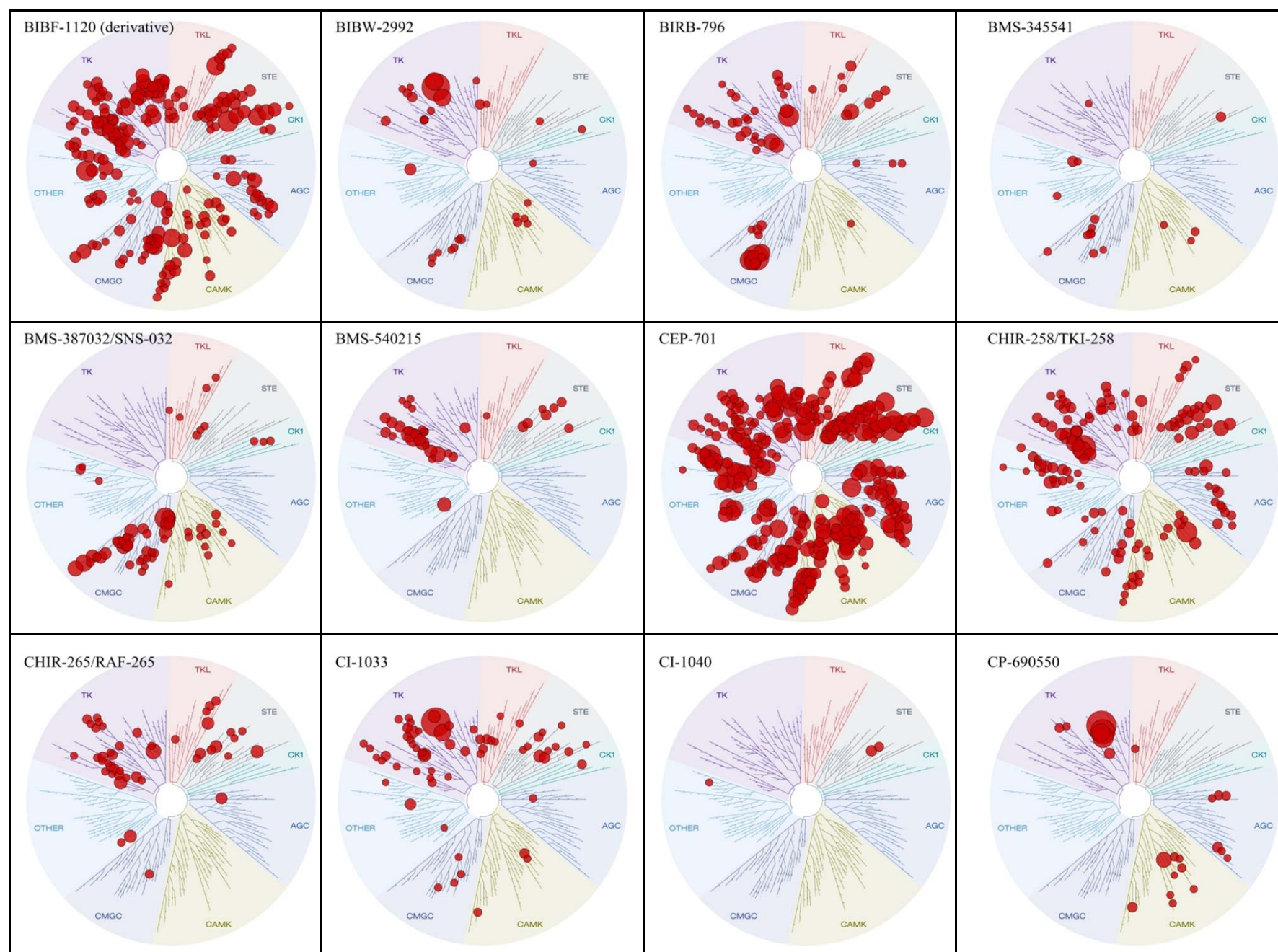


Supplementary Figure 1. Comparison of K_d values reported here and published biochemical $IC_{50}/K_i/K_d$ values for interactions between inhibitors and their primary, intended targets. Values used to create this plot are reported in Supplementary Table 3. Comparative data were available for 66 of the 72 compounds addressed in this study. The line of equivalence is indicated in solid red, and dashed red lines indicate 10-fold offsets.

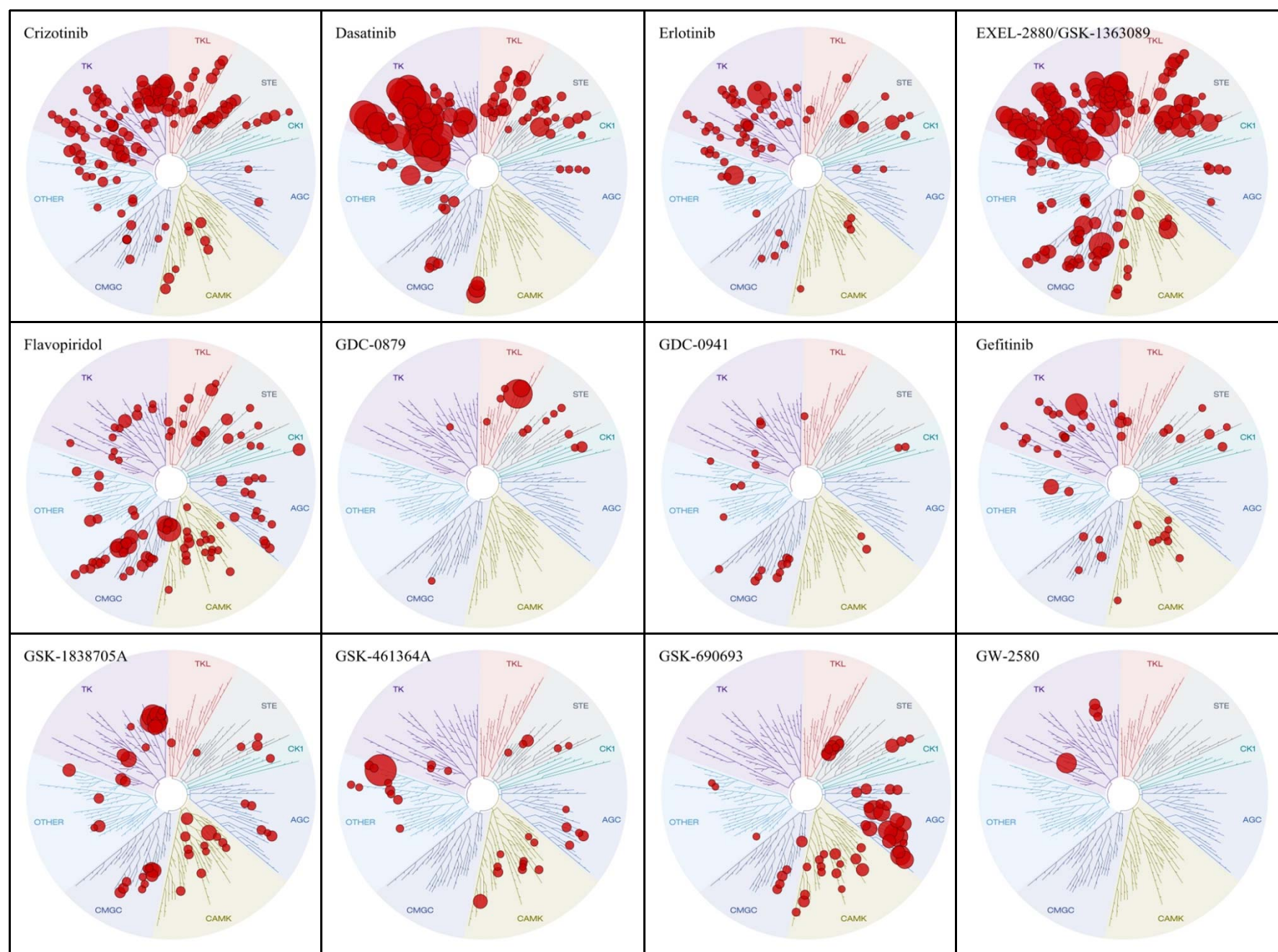
Supplementary Figure 2.



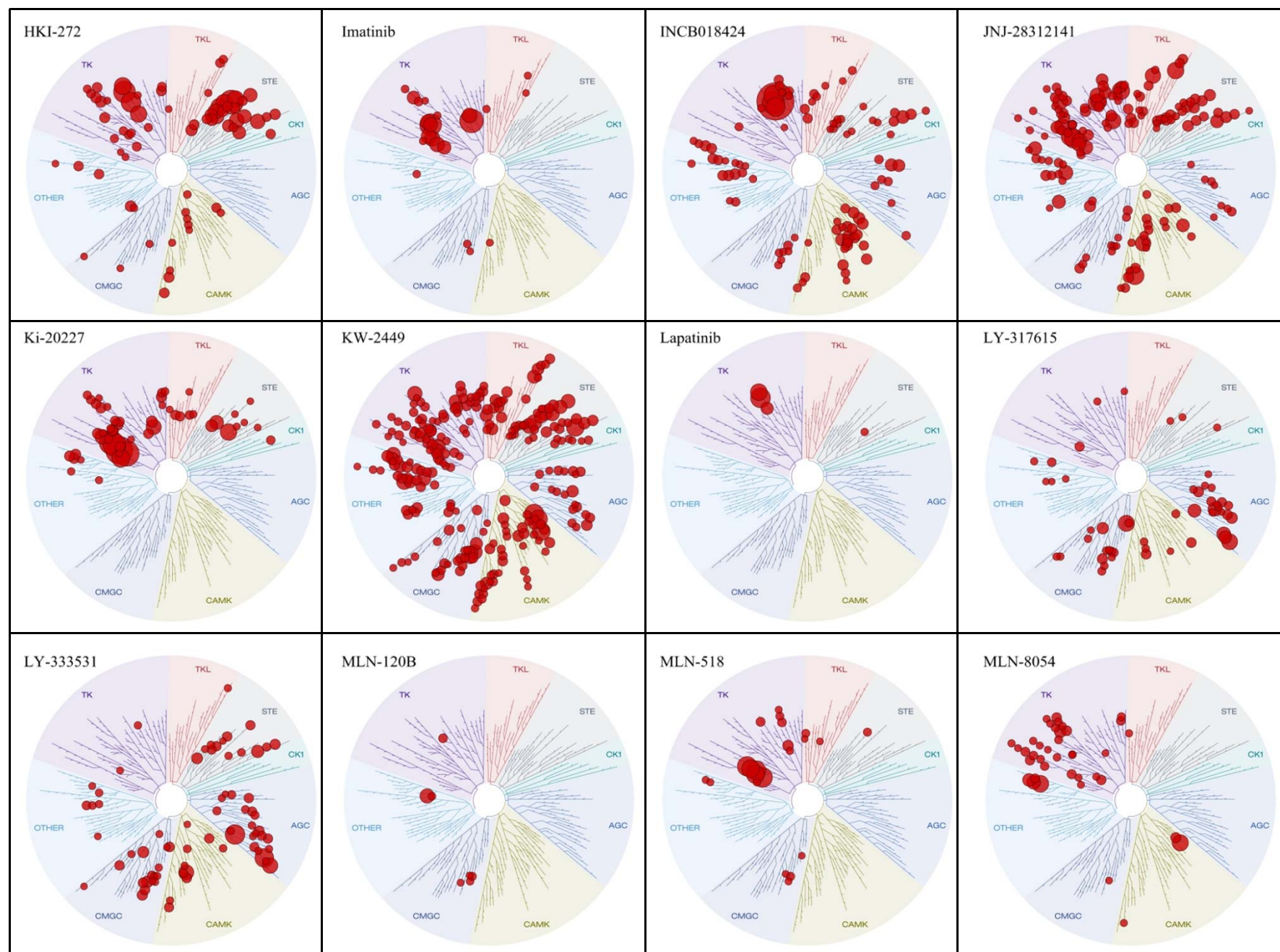
Supplementary Figure 2.



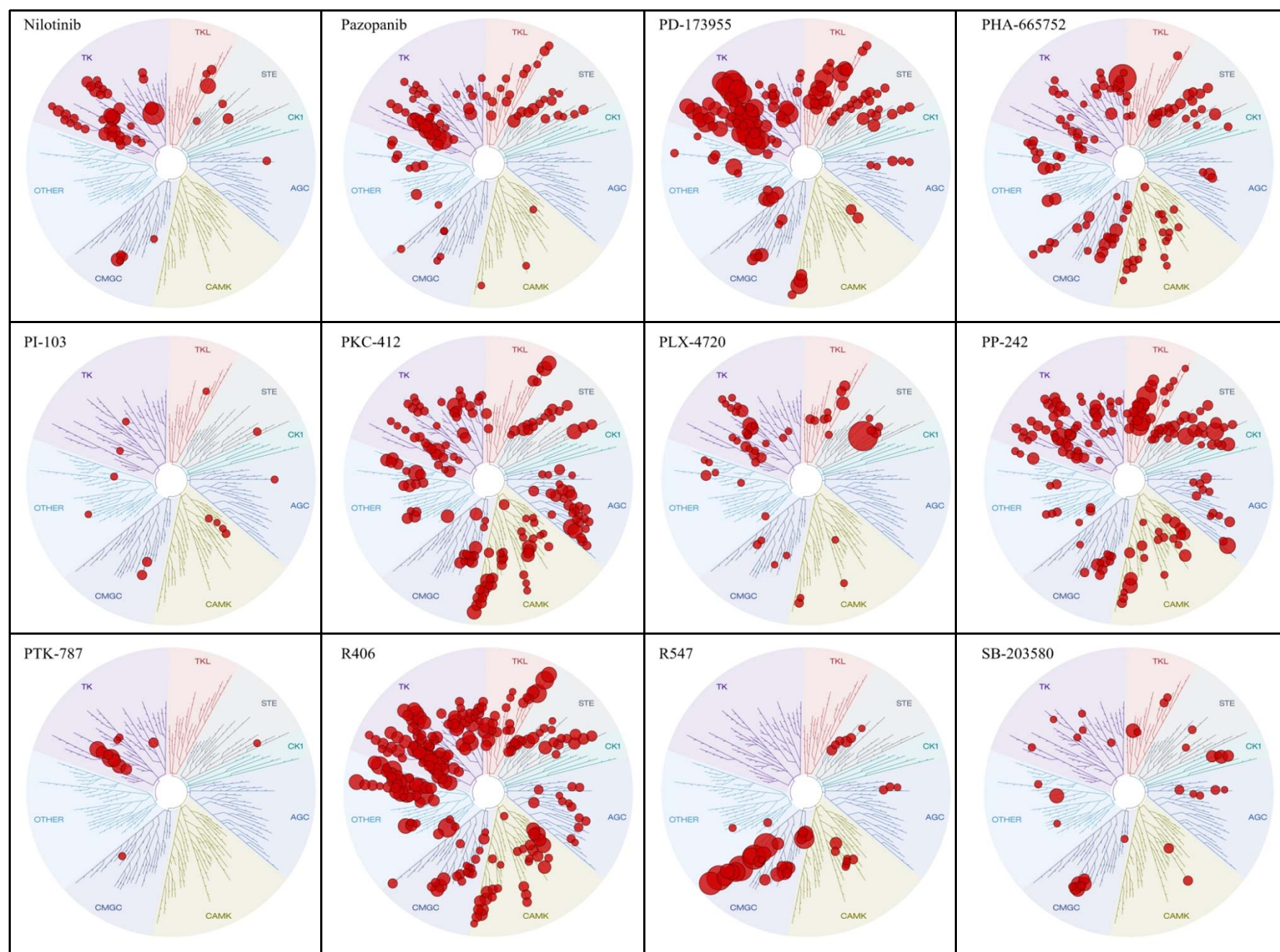
Supplementary Figure 2.



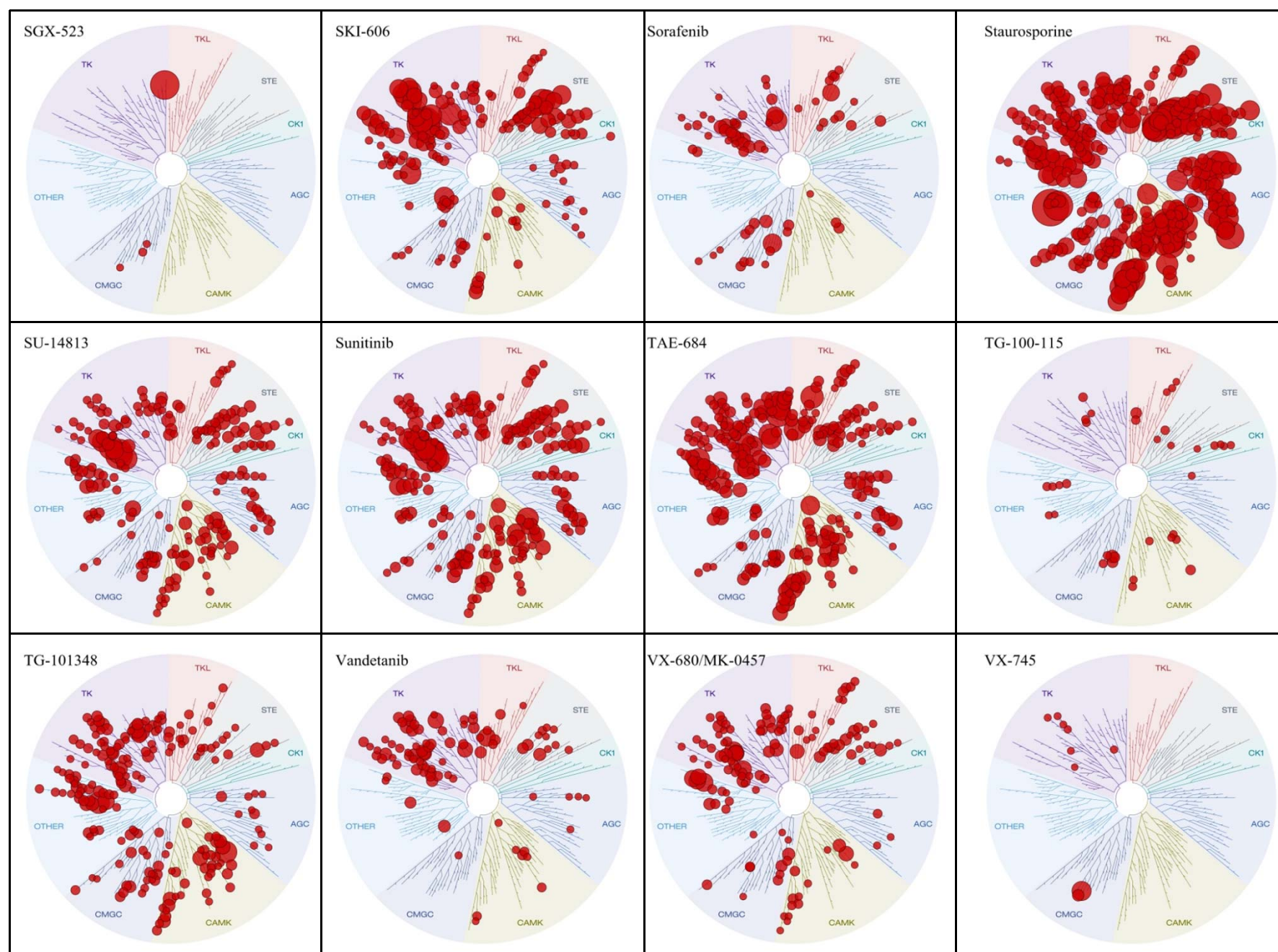
Supplementary Figure 2.



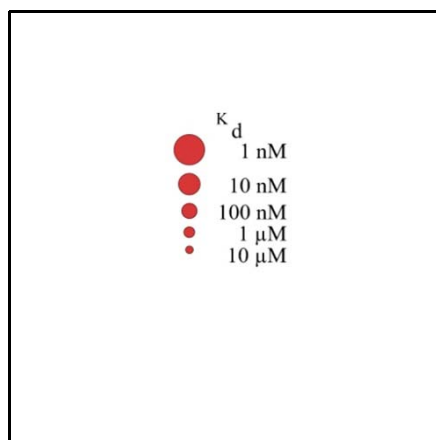
Supplementary Figure 2.



Supplementary Figure 2.

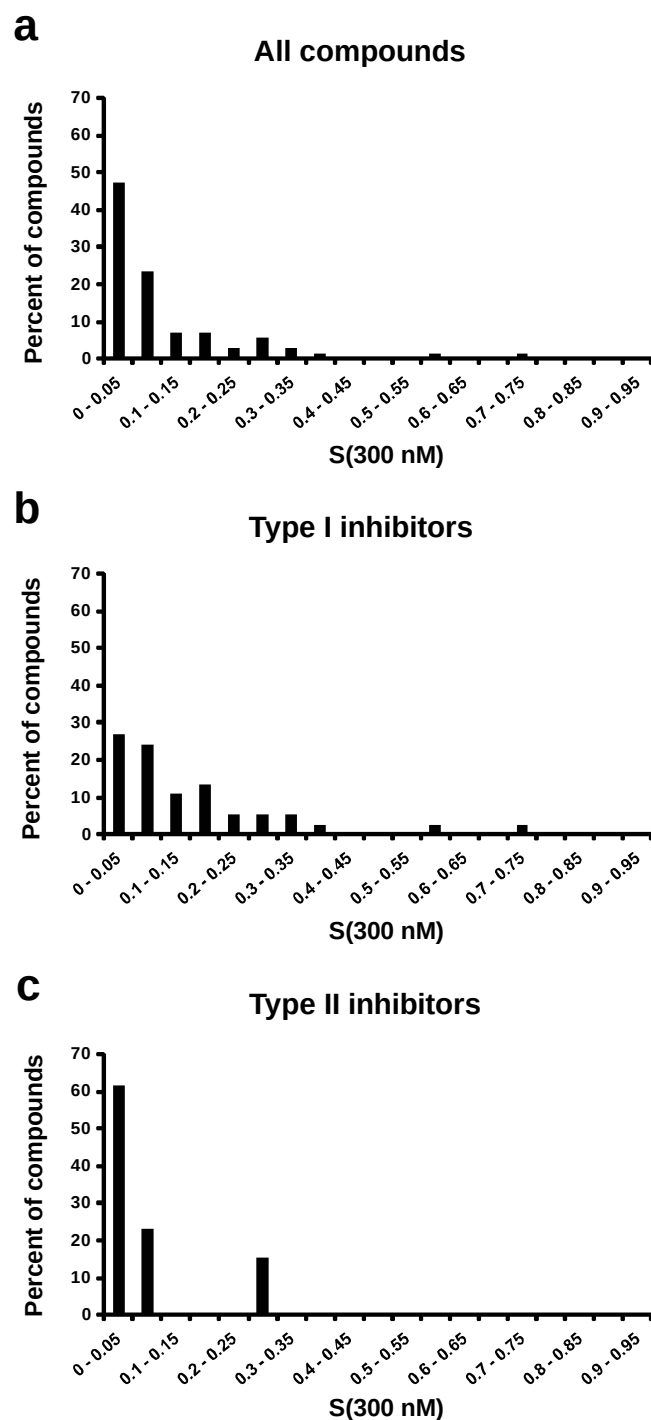


Supplementary Figure 2.



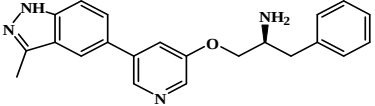
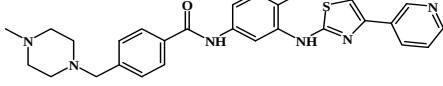
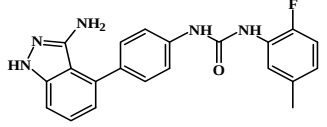
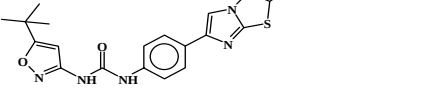
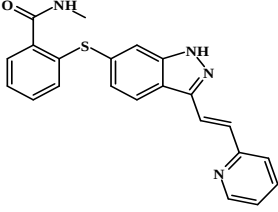
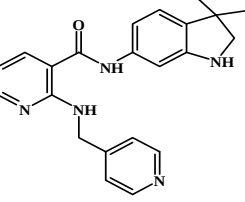
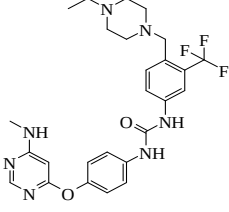
Supplementary Figure 2. Kinome interaction maps for the 72 compounds tested. Each red circle indicates a kinase found to bind to a compound. Larger circles indicate higher affinity interactions. Interactions with $K_d < 3 \mu\text{M}$ are shown.

Supplementary Figure 3.

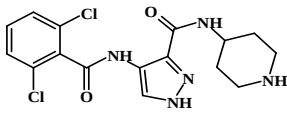
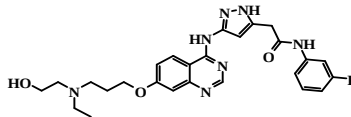
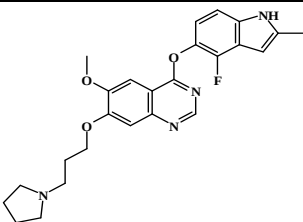
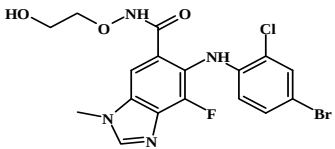
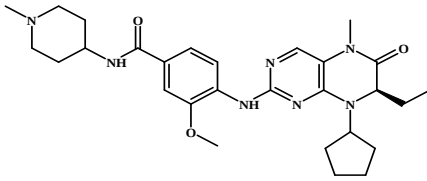
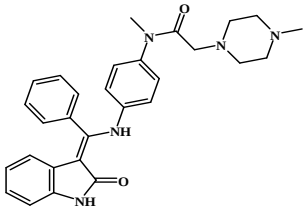
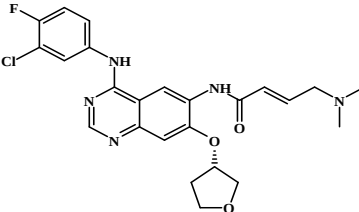


Supplementary Figure 3. Quantitative distribution of kinome-wide selectivity of compounds. An alternative analysis of the data shown in Figure 1 using a cutoff of S(300 nM) instead of S(3 μ M) and reduced bin sizes (0.05 units instead of 0.1). (a), (b), and (c) are otherwise as described in Figure 1.

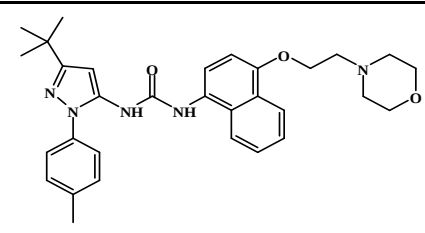
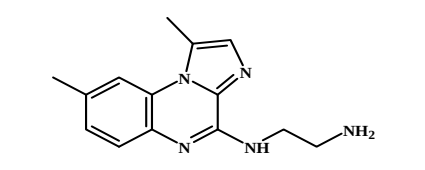
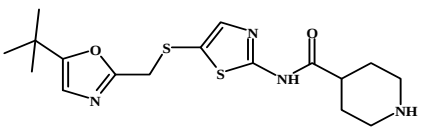
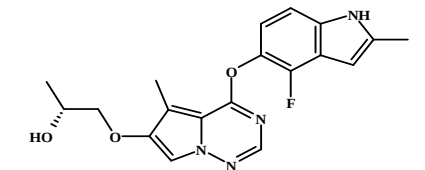
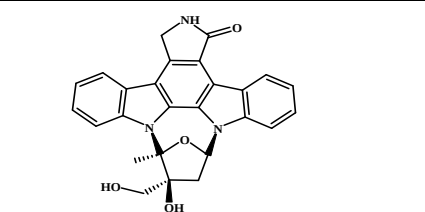
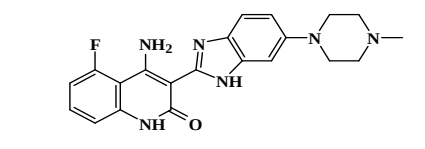
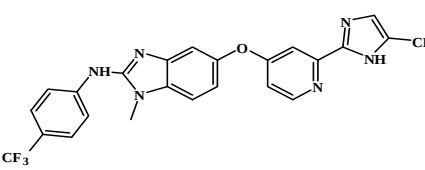
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
A-674563		CLK2	0.51
		CLK4	0.53
		CLK1	1.4
		DYRK1A	2.1
		DYRK1B	3.9
		PRKCH	9.5
		PRKCE	11
		PRKCQ	12
		CIT	13
		PRKG2	19
AB-1010		ABL1-nonphosphorylated	2.1
		CSF1R	7.6
		KIT	8.1
		PDGFRB	8.4
		DDR1	8.7
		PDGFRA	25
		DDR2	26
		LCK	31
		ABL1-phosphorylated	55
		LYN	61
ABT-869		FLT3	0.63
		PDGFRB	1.9
		KIT	2
		CSF1R	3.4
		PDGFRA	4.2
		FLT1	7.5
		VEGFR2	8.1
		MUSK	10
		FLT4	16
		EPHB6	33
AC220		FLT3	1.3
		KIT	4.8
		PDGFRB	7.7
		RET	8
		CSF1R	10
		PDGFRA	11
		FLT1	41
		FLT4	41
		DDR1	81
		VEGFR2	87
AG-013736		PDGFRA	0.51
		PDGFRB	0.57
		AURKC	1.3
		KIT	3.2
		FLT1	5.8
		VEGFR2	5.9
		AURKB	11
		PLK4	16
		CSF1R	21
		ABL1-phosphorylated	36
AMG-706		KIT	3.7
		CSF1R	5.6
		ZAK	8
		PDGFRB	9.1
		FLT4	9.7
		PDGFRA	10
		FLT1	12
		RET	14
		VEGFR2	26
		FLT3	71
AST-487		TIE1	0.29
		CDKL2	0.52
		ABL1-nonphosphorylated	0.68
		DDR1	0.69
		FLT3	0.79
		LOK	0.92
		CDC2L5	0.94
		CDK8	1.4
		CDK11	1.5
		TAK1	1.5

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
AT-7519		PCTK2	0.95
		PCTK1	1.1
		CDKL5	2.6
		CDK7	2.8
		CDC2L5	3.2
		CDC2L2	5.2
		CDK9	5.8
		ICK	8.3
		CDC2L1	8.4
		GSK3B	11
AZD-1152HQA		MEK5	2.8
		AURKC	4.4
		AURKB	4.6
		FLT3	8.2
		KIT	17
		PDGFRA	38
		PDGFRB	41
		EPHB6	50
		RET	80
		HIPK4	97
AZD-2171		PDGFRB	0.32
		KIT	0.38
		PDGFRA	0.41
		FLT1	0.74
		VEGFR2	1.1
		DDR1	1.7
		FLT4	4.3
		STK35	5.4
		RET	6.1
		CSF1R	13
AZD-6244/ARRY-886		MEK1	99
		MEK2	530
		EGFR	7000
BI-2536		PLK1	0.19
		PLK2	0.81
		PLK3	4
		RPS6KA4(Kin.Dom.2-C-terminal)	12
		CAMKK1	22
		CAMKK2	23
		MYLK	97
		PIP5K2C	110
		DAPK3	130
		FAK	150
BIBF-1120 (derivative)		MEK5	1.8
		BIKE	2.2
		VEGFR2	2.9
		PKNB(M.tuberculosis)	3.6
		FLT3	3.8
		TAK1	4.1
		TRKA	4.5
		JAK1(JH2domain-pseudokinase)	4.8
		MELK	4.9
		YSK4	5.2
BIBW-2992		EGFR	0.25
		ERBB2	5
		ERBB4	6.3
		GAK	79
		BLK	220
		IRAK1	240
		EPHA6	340
		HIPK4	360
		PHKG2	470
		ABL1-phosphorylated	570

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
BIRB-796		p38-alpha	0.45
		DDR1	1.9
		p38-gamma	2.9
		p38-beta	7.2
		JNK2	7.3
		TIE1	8.3
		LOK	12
		TIE2	20
		DDR2	33
		p38-delta	78
		IKK-beta	130
		YSK4	260
BMS-345541		CDC2L2	390
		CDC2L1	420
		ERK5	620
		CDK7	680
		MYLK4	700
		CDC2L5	800
		PCTK1	890
		ERN1	1000
		CDKL5	1.7
		PCTK1	7.1
		PCTK2	13
		CDC2L5	23
BMS-387032/SNS-032		GSK3A	28
		CDK7	31
		GSK3B	37
		CDKL2	41
		PCTK3	44
		CDC2L2	48
		VEGFR2	5
		FLT1	10
		PDGFRA	11
		STK35	26
		KIT	36
		PDGFRB	50
BMS-540215		FLT4	56
		FGFR1	99
		FGFR2	110
		DDR1	160
		PHKG1	0.39
		YSK4	0.52
		LATS2	1
		SNARK	1
		MKNK2	1.4
		PLK4	1.5
		IRAK4	1.7
		PHKG2	1.7
CEP-701		PKN2	1.8
		JAK3(JH1domain-catalytic)	2.3
		FLT3	0.64
		MLCK	2
		PDGFRB	3.8
		KIT	7.5
		MINK	9.1
		YSK4	12
		TNIK	24
		MEK5	32
		MAST1	40
		HPK1	44
CHIR-258/TKI-258		DDR1	13
		EPHB6	43
		YSK4	54
		GCN2(Kin.Dom.2,S808G)	55
		LOK	60
		ZAK	63
		CIT	87
		TAOK2	140
		RET	150
		TIE1	150
CHIR-265/RAF-265			

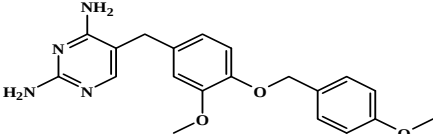
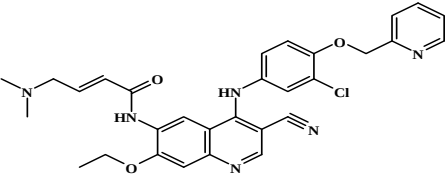
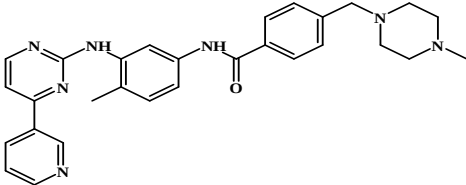
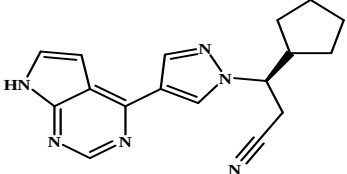
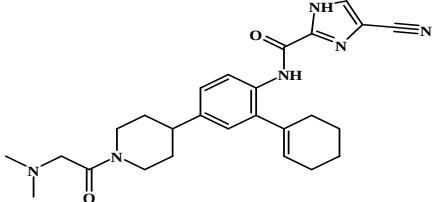
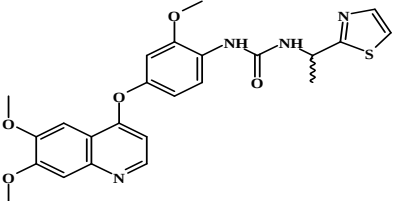
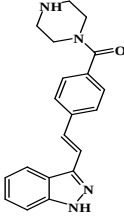
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
CI-1033		EGFR	0.19
		ERBB4	6.5
		ABL1-phosphorylated	30
		BLK	45
		ERBB2	56
		MEK5	60
		GAK	100
		MKK7	110
		ABL1-nonphosphorylated	210
		ERBB3	210
CI-1040		MEK1	120
		MEK2	370
		AURKC	1800
		PDGFRB	3100
		CAMK2A	3400
CP-690550		JAK3(JH1domain-catalytic)	0.16
		JAK2(JH1domain-catalytic)	0.58
		JAK1(JH1domain-catalytic)	1.6
		TYK2(JH1domain-catalytic)	4.8
		DCAMKL3	12
		TNK1	120
		PKN1	170
		SNARK	240
		ROCK2	420
		LCK	460
Crizotinib		MET	2.1
		ALK	3.3
		MERTK	3.6
		ROS1	4.1
		EPHB6	6
		AXL	7.8
		LTK	12
		SLK	18
		MST1R	25
		LCK	30
Dasatinib		ABL1-nonphosphorylated	0.029
		EPHB6	0.039
		ABL1-phosphorylated	0.046
		EPHA3	0.093
		ABL2	0.17
		LCK	0.2
		BLK	0.21
		SRC	0.21
		EPHA5	0.24
		EPHA8	0.24
Erlotinib		EGFR	0.67
		GAK	3.1
		LOK	19
		YSK4	25
		SLK	26
		ABL1-phosphorylated	76
		MEK5	96
		BLK	190
		ABL2	200
		ERBB4	230
EXEL-2880/GSK-1363089		AXL	0.093
		DDR1	0.2
		MERTK	0.27
		HIPK4	0.51
		LOK	0.53
		RET	0.74
		TIE1	0.79
		FLT3	0.9
		PDGFRB	0.96
		EPHA3	1

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Flavopiridol		ICK	0.69
		CDK4-cyclinD3	3.3
		CDK9	6.4
		CDKL5	7.1
		CDK4-cyclinD1	9
		CDK7	23
		MAK	28
		TYK2(JH2domain-pseudokinase)	35
		TNNI3K	55
		CDK11	57
GDC-0879		BRAF	0.19
		RAF1	6.6
		CSNK1E	130
		YSK4	910
		MINK	1000
		RIOK2	1200
		LOK	1300
		SLK	1300
		CSNK1D	1400
		BMPR1B	1800
GDC-0941		PIK3CA	1.1
		PIK3CD	5
		PIK3CB	16
		PIK3CG	48
		PIK3C2B	130
		MTOR	200
		PIK3C2G	300
		JAK1(JH2domain-pseudokinase)	430
		HIPK2	520
		JNK3	560
Gefitinib		EGFR	1
		GAK	13
		IRAK1	69
		YSK4	240
		MKNK1	290
		HIPK4	310
		ERBB4	410
		CSNK1E	430
		LOK	470
		ABL1-phosphorylated	480
GSK-1838705A		ALK	0.55
		LTK	1.1
		INSR	1.7
		IGF1R	7
		INSRR	8.6
		FER	9.3
		MYLK	11
		ROS1	15
		CLK2	16
		CLK1	21
GSK-461364A		PLK1	0.094
		SNARK	23
		LOK	69
		RSK2(Kin.Dom.1-N-terminal)	190
		PIM1	250
		NEK2	260
		RSK4(Kin.Dom.1-N-terminal)	350
		BIKE	470
		PLK2	500
		CAMK2D	620
GSK-690693		AKT2	2.1
		AKT1	2.2
		PRKCH	2.4
		AKT3	3
		PRKG2	3.1
		PKNB(M.tuberculosis)	3.2
		PRKCE	5.3
		PRKX	7.2
		PAK7	8.9
		PKAC-beta	13

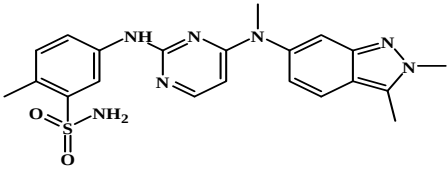
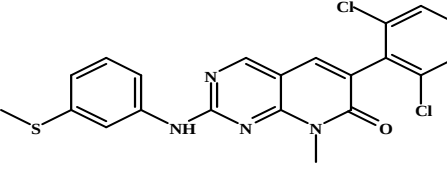
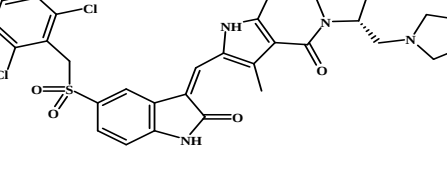
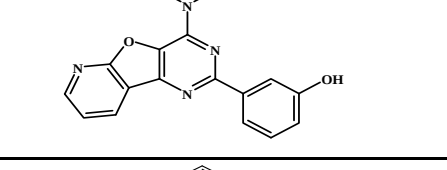
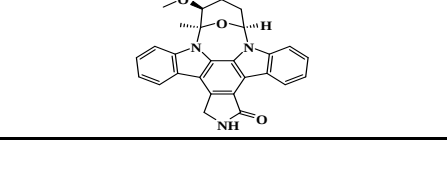
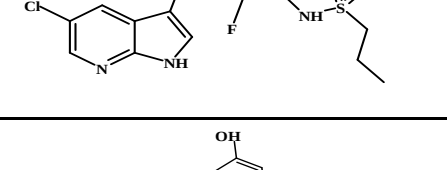
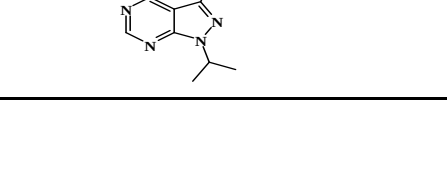
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
GW-2580		CSF1R	2.2
		TRKB	36
		TRKC	120
		TRKA	630
HKI-272		MAP4K5	0.65
		EGFR	1.1
		ERBB4	2.4
		ERBB2	6
		MST3	6.5
		MST4	7.4
		ERBB3	7.7
		MAP4K3	7.7
		YSK1	12
		LOK	13
		DDR1	0.7
Imatinib		ABL1-nonphosphorylated	1.1
		ABL2	10
		CSF1R	11
		KIT	13
		PDGFRB	14
		DDR2	15
		ABL1-phosphorylated	21
		PDGFRA	31
		LCK	40
INC018424		JAK2(JH1domain-catalytic)	0.036
		TYK2(JH1domain-catalytic)	0.9
		JAK3(JH1domain-catalytic)	2
		JAK1(JH1domain-catalytic)	3.4
		MAP3K2	41
		CAMK2A	46
		ROCK2	52
		ROCK1	60
		DCAMKL1	68
		DAPK1	72
JNJ-28312141		CSF1R	3.2
		KIT	3.6
		AMPK-alpha2	4.1
		AXL	5.3
		TYK2(JH1domain-catalytic)	5.8
		TAK1	7.2
		RET	9.2
		DDR1	9.4
		CHEK1	9.9
		JAK2(JH1domain-catalytic)	9.9
Ki-20227		PDGFRB	0.29
		PDGFRA	0.49
		KIT	0.69
		CSF1R	0.83
		EPHB6	5
		MEK5	7.4
		DDR1	12
		VEGFR2	18
		LOK	22
		SLK	60
KW-2449		DRAK1	2.9
		MLCK	4.3
		DRAK2	4.8
		YSK4	5.2
		BIKE	9.6
		MAP4K2	11
		LOK	13
		SLK	13
		FLT3	15
		SRPK2	15

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM	
Lapatinib		EGFR	2.4	
		ERBB2	7	
		ERBB4	54	
		PIK3C2B	670	
		PIK4CB	940	
		MEK5	1100	
		SLK	3300	
		RIPK2	3600	
		LOK	4400	
		MKK7	4400	
LY-317615		GSK3B	8.3	
		PRKCE	8.9	
		PRKCD	25	
		RSK4(Kin.Dom.1-N-terminal)	25	
		PRKCQ	36	
		PRKCH	46	
		ERK8	76	
		RSK2(Kin.Dom.1-N-terminal)	87	
		DYRK1A	160	
		PRKG2	170	
LY-333531		PRKCQ	2.5	
		PRKCD	3.6	
		PRKCE	11	
		PIM3	12	
		YSK4	48	
		HIPK3	77	
		ERK8	88	
		TAOK3	97	
		HIPK2	140	
		MAP3K3	170	
MLN-120B		IKK-beta	19	
		PIK4CB	270	
		TYK2(JH2domain-pseudokinase)	500	
		CLK1	640	
		CLK4	1000	
		IKK-alpha	1000	
		DYRK1A	2100	
		CLK2	2300	
MLN-518		PDGFRA	2.4	
		KIT	2.7	
		FLT3	3	
		PDGFRB	4.5	
		CSF1R	4.9	
		DDR2	120	
		EGFR	410	
		TRKA	450	
		CLK1	630	
		IRAK3	730	
MLN-8054		AURKA	6.5	
		DRAK2	8.1	
		AURKC	26	
		AURKB	43	
		BLK	68	
		DRAK1	190	
		FGR	220	
		YES	260	
		TIE2	300	
		EPHB1	330	
Nilotinib		DDR1	1.1	
		ABL1-nonphosphorylated	10	
		ZAK	11	
		ABL1-phosphorylated	13	
		ABL2	26	
		KIT	29	
		DDR2	33	
		p38-beta	36	
		EPHA8	37	
		CSF1R	45	

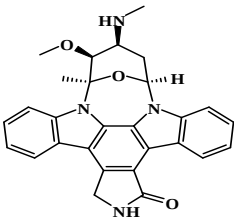
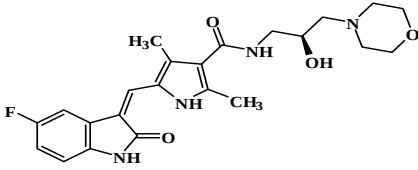
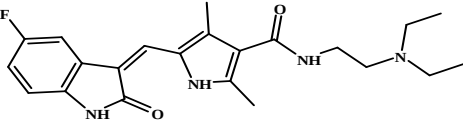
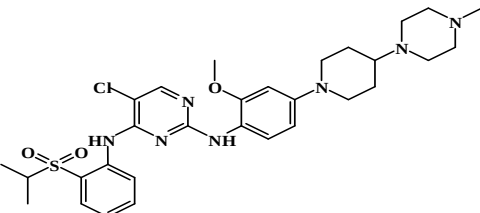
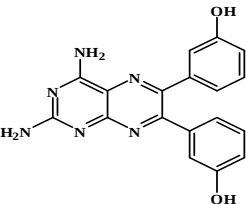
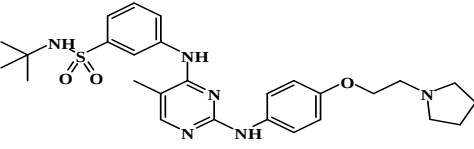
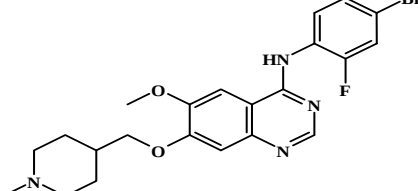
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Pazopanib		PDGFRB	2
		KIT	2.8
		PDGFRA	4.9
		CSF1R	7.9
		FLT1	14
		VEGFR2	14
		FLT4	27
		TAOK3	45
		DDR1	57
		EPHB6	81
PD-173955		ABL1-phosphorylated	0.58
		CSF1R	0.67
		ABL2	0.69
		SRC	0.71
		LCK	1.1
		PDGFRB	1.4
		ABL1-nonphosphorylated	1.5
		BLK	1.5
		KIT	1.8
		EPHB6	2
PHA-665752		MET	0.27
		SRPK1	41
		TAOK3	43
		BIKE	47
		GRK7	61
		HIPK3	63
		CAMKK1	64
		DDR1	70
		CAMKK2	73
		YSK4	78
PI-103		PIK3CA	1.5
		PIK3CB	1.7
		PIK3C2B	10
		MTOR	12
		PIK3CG	16
		PIK3CD	17
		PIK3C2G	41
		HIPK2	290
		HIPK3	310
		PIP5K2C	620
PKC-412		PKN1	9.3
		TBK1	9.3
		FLT3	11
		JAK3(JH1domain-catalytic)	12
		MLK1	15
		PKN2	15
		YSK4	15
		MLK3	17
		CAMK2A	20
		MARK3	21
PLX-4720		MEK5	0.16
		PFCDPK1(P.falciparum)	1.7
		SRMS	21
		ZAK	41
		BRK	48
		FGR	62
		RAF1	170
		KIT	180
		MEK4	190
		NEK11	190
PP-242		BMPR1B	2.1
		ACVRL1	2.9
		MTOR	3
		ACVR1	4
		RET	4.8
		YSK4	5.1
		MEK5	7.3
		ACVR2B	7.6
		PRKCE	9
		JAK2(JH1domain-catalytic)	11

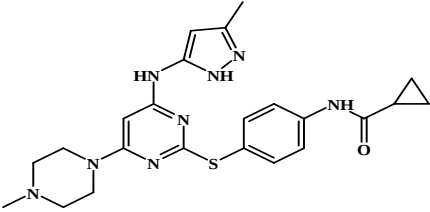
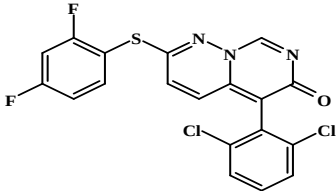
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
PTK-787		KIT	5.1
		FLT1	9.6
		PDGFRB	25
		CSF1R	45
		VEGFR2	62
		PDGFRA	96
		DDR1	270
		FLT4	330
		CDK11	1500
		FRK	1800
R406		FLT3	0.71
		STK16	1.7
		GCN2(Kin.Dom.2,S808G)	3.3
		PDGFRB	3.3
		JAK2(JH1domain-catalytic)	3.5
		MLK2	3.8
		RET	4.1
		PLK4	4.2
		MLK1	4.3
		PLK3	5.1
R547		CDK2	0.53
		PCTK1	0.54
		CDK7	0.58
		CDK4-cyclinD1	0.61
		CDK4-cyclinD3	0.81
		PCTK2	0.86
		ICK	2.2
		CDK3	3.2
		PFTAIRE2	7.2
		CDK5	7.4
SB-203580		p38-alpha	12
		GAK	19
		RIPK2	24
		NLK	25
		JNK3	35
		CSNK1D	37
		p38-beta	70
		CSNK1A1	75
		CSNK1E	100
		JNK2	130
SGX-523		MET	0.19
		DYRK1A	780
		DYRK1B	1800
		JNK3	1900
		PIP5K2C	3300
		JNK1	4200
		YSK4	6400
SKI-606		ABL1-phosphorylated	0.057
		ABL1-nonphosphorylated	0.12
		MAP4K5	0.5
		LCK	0.59
		ERBB3	0.77
		SRC	1
		GAK	1.3
		FRK	1.4
		ABL2	1.5
		STK35	2
Sorafenib		DDR1	1.5
		HIPK4	3.3
		ZAK	6.3
		DDR2	6.6
		FLT3	13
		RET	13
		CSF1R	28
		KIT	28
		FLT1	31
		PDGFRB	37

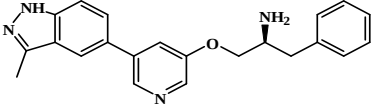
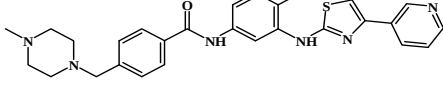
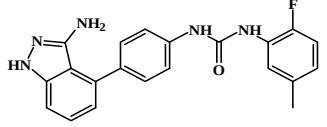
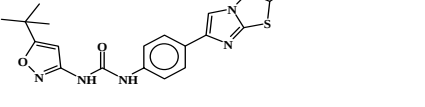
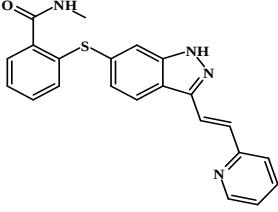
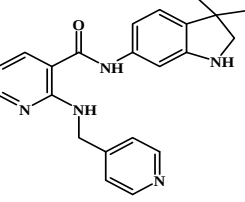
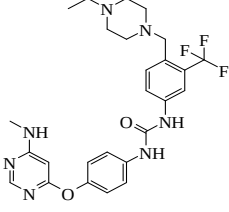
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Staurosporine		SLK	0.024
		LOK	0.037
		CAMKK1	0.039
		SNARK	0.086
		PHKG2	0.14
		CAMK2A	0.16
		CAMKK2	0.16
		MST2	0.18
		MST1	0.19
		TAOK3	0.22
SU-14813		PDGFRB	0.29
		FLT3	0.54
		KIT	0.68
		PDGFRA	1.1
		VEGFR2	2.3
		CSF1R	3.6
		HUNK	3.7
		FLT1	4.7
		STK35	8.2
		YSK4	12
Sunitinib		PDGFRB	0.075
		KIT	0.37
		FLT3	0.41
		PDGFRA	0.79
		DRAK1	1
		VEGFR2	1.5
		FLT1	1.8
		CSF1R	2.5
		BIKE	5.5
		PHKG1	5.5
TAE-684		ROS1	0.49
		ULK1	0.83
		BMPR1B	0.85
		PLK4	0.93
		LTK	0.95
		ALK	1.1
		FAK	1.1
		PYK2	1.1
		SNARK	1.2
		FER	1.4
TG-100-115		PIK3C2G	3.2
		PIK3CG	5.3
		PIK3C2B	7.3
		TRPM6	7.9
		CLK2	43
		PIK3CA	59
		PIK3CB	80
		ADCK3	94
		RIPK4	97
		CLK4	130
TG-101348		GAK	1.1
		JAK2(JH1domain-catalytic)	1.1
		DAPK3	1.2
		STK16	6.6
		DCAMKL3	13
		FLT3	13
		DAPK1	16
		JAK1(JH1domain-catalytic)	18
		YSK4	19
		TYK2(JH1domain-catalytic)	21
Vandetanib		RIPK2	4.6
		EGFR	9.5
		DDR1	11
		ABL1-phosphorylated	16
		LCK	17
		RET	34
		ABL1-nonphosphorylated	48
		MEK5	49
		EPHA6	50
		STK35	56

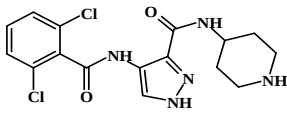
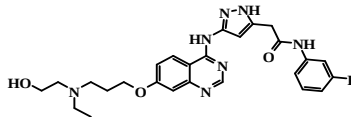
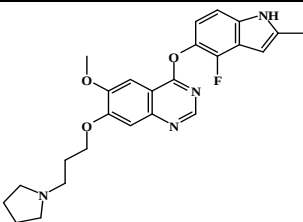
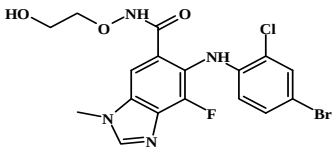
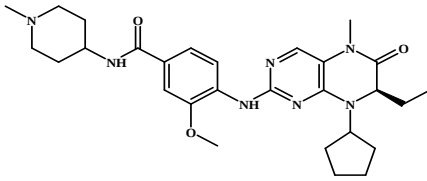
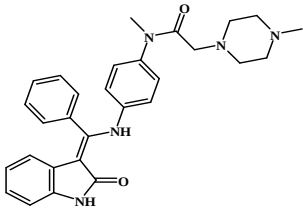
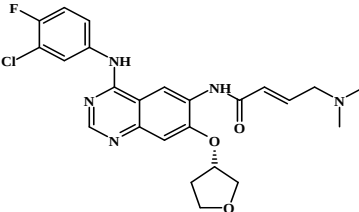
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
VX-680/MK-0457		AURKA	3.9
		ABL2	4
		AURKC	6.3
		FLT3	6.5
		AURKB	7.4
		ABL1-phosphorylated	7.5
		PLK4	9.2
		ABL1-nonphosphorylated	13
		MLCK	15
		RIPK1	20
VX-745		p38-alpha	2.8
		p38-beta	74
		DDR1	1100
		FGR	1300
		YES	1600
		LYN	1700
		ABL2	1900
		FYN	2100
		CSF1R	2600
		BLK	3100

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
A-674563		CLK2	0.51
		CLK4	0.53
		CLK1	1.4
		DYRK1A	2.1
		DYRK1B	3.9
		PRKCH	9.5
		PRKCE	11
		PRKCQ	12
		CIT	13
		PRKG2	19
AB-1010		ABL1-nonphosphorylated	2.1
		CSF1R	7.6
		KIT	8.1
		PDGFRB	8.4
		DDR1	8.7
		PDGFRA	25
		DDR2	26
		LCK	31
		ABL1-phosphorylated	55
		LYN	61
ABT-869		FLT3	0.63
		PDGFRB	1.9
		KIT	2
		CSF1R	3.4
		PDGFRA	4.2
		FLT1	7.5
		VEGFR2	8.1
		MUSK	10
		FLT4	16
		EPHB6	33
AC220		FLT3	1.3
		KIT	4.8
		PDGFRB	7.7
		RET	8
		CSF1R	10
		PDGFRA	11
		FLT1	41
		FLT4	41
		DDR1	81
		VEGFR2	87
AG-013736		PDGFRA	0.51
		PDGFRB	0.57
		AURKC	1.3
		KIT	3.2
		FLT1	5.8
		VEGFR2	5.9
		AURKB	11
		PLK4	16
		CSF1R	21
		ABL1-phosphorylated	36
AMG-706		KIT	3.7
		CSF1R	5.6
		ZAK	8
		PDGFRB	9.1
		FLT4	9.7
		PDGFRA	10
		FLT1	12
		RET	14
		VEGFR2	26
		FLT3	71
AST-487		TIE1	0.29
		CDKL2	0.52
		ABL1-nonphosphorylated	0.68
		DDR1	0.69
		FLT3	0.79
		LOK	0.92
		CDC2L5	0.94
		CDK8	1.4
		CDK11	1.5
		TAK1	1.5

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
AT-7519		PCTK2	0.95
		PCTK1	1.1
		CDKL5	2.6
		CDK7	2.8
		CDC2L5	3.2
		CDC2L2	5.2
		CDK9	5.8
		ICK	8.3
		CDC2L1	8.4
		GSK3B	11
AZD-1152HQA		MEK5	2.8
		AURKC	4.4
		AURKB	4.6
		FLT3	8.2
		KIT	17
		PDGFRA	38
		PDGFRB	41
		EPHB6	50
		RET	80
		HIPK4	97
AZD-2171		PDGFRB	0.32
		KIT	0.38
		PDGFRA	0.41
		FLT1	0.74
		VEGFR2	1.1
		DDR1	1.7
		FLT4	4.3
		STK35	5.4
		RET	6.1
		CSF1R	13
AZD-6244/ARRY-886		MEK1	99
		MEK2	530
		EGFR	7000
BI-2536		PLK1	0.19
		PLK2	0.81
		PLK3	4
		RPS6KA4(Kin.Dom.2-C-terminal)	12
		CAMKK1	22
		CAMKK2	23
		MYLK	97
		PIP5K2C	110
		DAPK3	130
		FAK	150
BIBF-1120 (derivative)		MEK5	1.8
		BIKE	2.2
		VEGFR2	2.9
		PKNB(M.tuberculosis)	3.6
		FLT3	3.8
		TAK1	4.1
		TRKA	4.5
		JAK1(JH2domain-pseudokinase)	4.8
		MELK	4.9
		YSK4	5.2
BIBW-2992		EGFR	0.25
		ERBB2	5
		ERBB4	6.3
		GAK	79
		BLK	220
		IRAK1	240
		EPHA6	340
		HIPK4	360
		PHKG2	470
		ABL1-phosphorylated	570

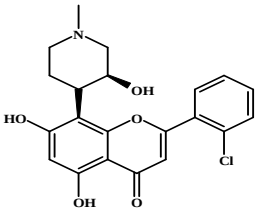
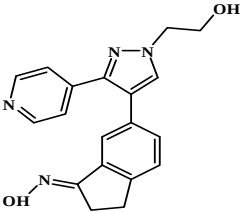
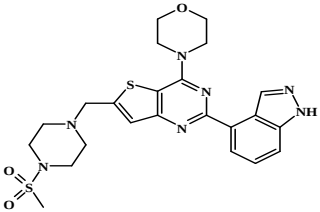
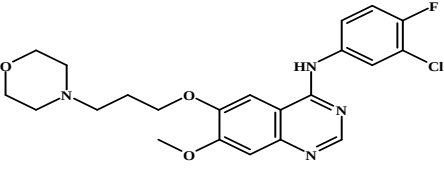
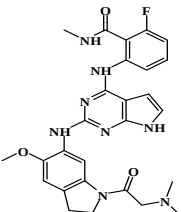
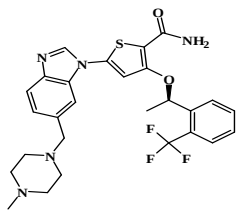
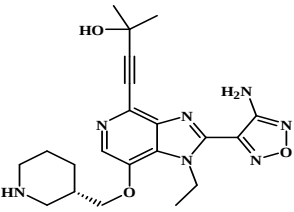
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
BIRB-796		p38-alpha	0.45
		DDR1	1.9
		p38-gamma	2.9
		p38-beta	7.2
		JNK2	7.3
		TIE1	8.3
		LOK	12
		TIE2	20
		DDR2	33
		p38-delta	78
		IKK-beta	130
		YSK4	260
BMS-345541		CDC2L2	390
		CDC2L1	420
		ERK5	620
		CDK7	680
		MYLK4	700
		CDC2L5	800
		PCTK1	890
		ERN1	1000
		CDKL5	1.7
		PCTK1	7.1
		PCTK2	13
		CDC2L5	23
BMS-387032/SNS-032		GSK3A	28
		CDK7	31
		GSK3B	37
		CDKL2	41
		PCTK3	44
		CDC2L2	48
		VEGFR2	5
		FLT1	10
		PDGFRA	11
		STK35	26
		KIT	36
		PDGFRB	50
BMS-540215		FLT4	56
		FGFR1	99
		FGFR2	110
		DDR1	160
		PHKG1	0.39
		YSK4	0.52
		LATS2	1
		SNARK	1
		MKNK2	1.4
		PLK4	1.5
		IRAK4	1.7
		PHKG2	1.7
CEP-701		PKN2	1.8
		JAK3(JH1domain-catalytic)	2.3
		FLT3	0.64
		MLCK	2
		PDGFRB	3.8
		KIT	7.5
		MINK	9.1
		YSK4	12
		TNIK	24
		MEK5	32
		MAST1	40
		HPK1	44
CHIR-258/TKI-258		DDR1	13
		EPHB6	43
		YSK4	54
		GCN2(Kin.Dom.2,S808G)	55
		LOK	60
		ZAK	63
		CIT	87
		TAOK2	140
		RET	150
		TIE1	150
CHIR-265/RAF-265			

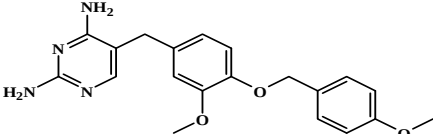
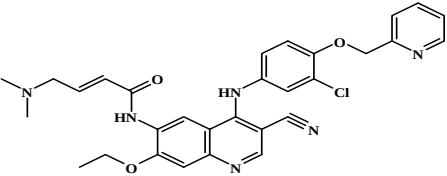
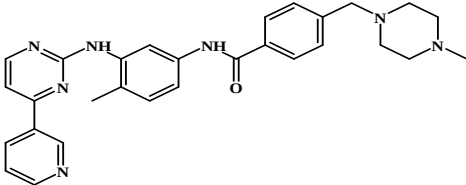
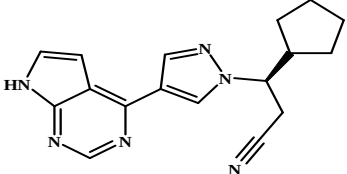
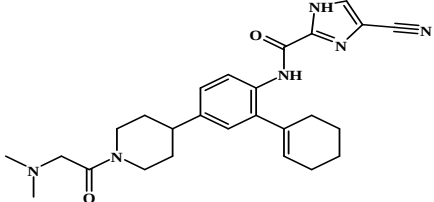
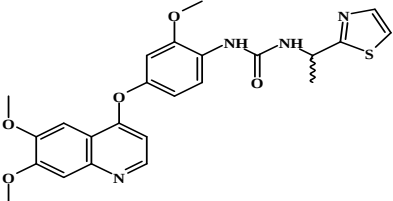
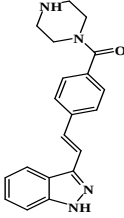
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
CI-1033		EGFR	0.19
		ERBB4	6.5
		ABL1-phosphorylated	30
		BLK	45
		ERBB2	56
		MEK5	60
		GAK	100
		MKK7	110
		ABL1-nonphosphorylated	210
		ERBB3	210
CI-1040		MEK1	120
		MEK2	370
		AURKC	1800
		PDGFRB	3100
		CAMK2A	3400
CP-690550		JAK3(JH1domain-catalytic)	0.16
		JAK2(JH1domain-catalytic)	0.58
		JAK1(JH1domain-catalytic)	1.6
		TYK2(JH1domain-catalytic)	4.8
		DCAMKL3	12
		TNK1	120
		PKN1	170
		SNARK	240
		ROCK2	420
		LCK	460
Crizotinib		MET	2.1
		ALK	3.3
		MERTK	3.6
		ROS1	4.1
		EPHB6	6
		AXL	7.8
		LTK	12
		SLK	18
		MST1R	25
		LCK	30
Dasatinib		ABL1-nonphosphorylated	0.029
		EPHB6	0.039
		ABL1-phosphorylated	0.046
		EPHA3	0.093
		ABL2	0.17
		LCK	0.2
		BLK	0.21
		SRC	0.21
		EPHA5	0.24
		EPHA8	0.24
Erlotinib		EGFR	0.67
		GAK	3.1
		LOK	19
		YSK4	25
		SLK	26
		ABL1-phosphorylated	76
		MEK5	96
		BLK	190
		ABL2	200
		ERBB4	230
EXEL-2880/GSK-1363089		AXL	0.093
		DDR1	0.2
		MERTK	0.27
		HIPK4	0.51
		LOK	0.53
		RET	0.74
		TIE1	0.79
		FLT3	0.9
		PDGFRB	0.96
		EPHA3	1

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Flavopiridol		ICK	0.69
		CDK4-cyclinD3	3.3
		CDK9	6.4
		CDKL5	7.1
		CDK4-cyclinD1	9
		CDK7	23
		MAK	28
		TYK2(JH2domain-pseudokinase)	35
		TNNI3K	55
		CDK11	57
GDC-0879		BRAF	0.19
		RAF1	6.6
		CSNK1E	130
		YSK4	910
		MINK	1000
		RIOK2	1200
		LOK	1300
		SLK	1300
		CSNK1D	1400
		BMPR1B	1800
GDC-0941		PIK3CA	1.1
		PIK3CD	5
		PIK3CB	16
		PIK3CG	48
		PIK3C2B	130
		MTOR	200
		PIK3C2G	300
		JAK1(JH2domain-pseudokinase)	430
		HIPK2	520
		JNK3	560
Gefitinib		EGFR	1
		GAK	13
		IRAK1	69
		YSK4	240
		MKNK1	290
		HIPK4	310
		ERBB4	410
		CSNK1E	430
		LOK	470
		ABL1-phosphorylated	480
GSK-1838705A		ALK	0.55
		LTK	1.1
		INSR	1.7
		IGF1R	7
		INSRR	8.6
		FER	9.3
		MYLK	11
		ROS1	15
		CLK2	16
		CLK1	21
GSK-461364A		PLK1	0.094
		SNARK	23
		LOK	69
		RSK2(Kin.Dom.1-N-terminal)	190
		PIM1	250
		NEK2	260
		RSK4(Kin.Dom.1-N-terminal)	350
		BIKE	470
		PLK2	500
		CAMK2D	620
GSK-690693		AKT2	2.1
		AKT1	2.2
		PRKCH	2.4
		AKT3	3
		PRKG2	3.1
		PKNB(M.tuberculosis)	3.2
		PRKCE	5.3
		PRKX	7.2
		PAK7	8.9
		PKAC-beta	13

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
GW-2580		CSF1R	2.2
		TRKB	36
		TRKC	120
		TRKA	630
HKI-272		MAP4K5	0.65
		EGFR	1.1
		ERBB4	2.4
		ERBB2	6
		MST3	6.5
		MST4	7.4
		ERBB3	7.7
		MAP4K3	7.7
		YSK1	12
		LOK	13
		DDR1	0.7
Imatinib		ABL1-nonphosphorylated	1.1
		ABL2	10
		CSF1R	11
		KIT	13
		PDGFRB	14
		DDR2	15
		ABL1-phosphorylated	21
		PDGFRA	31
		LCK	40
INC018424		JAK2(JH1domain-catalytic)	0.036
		TYK2(JH1domain-catalytic)	0.9
		JAK3(JH1domain-catalytic)	2
		JAK1(JH1domain-catalytic)	3.4
		MAP3K2	41
		CAMK2A	46
		ROCK2	52
		ROCK1	60
		DCAMKL1	68
		DAPK1	72
JNJ-28312141		CSF1R	3.2
		KIT	3.6
		AMPK-alpha2	4.1
		AXL	5.3
		TYK2(JH1domain-catalytic)	5.8
		TAK1	7.2
		RET	9.2
		DDR1	9.4
		CHEK1	9.9
		JAK2(JH1domain-catalytic)	9.9
Ki-20227		PDGFRB	0.29
		PDGFRA	0.49
		KIT	0.69
		CSF1R	0.83
		EPHB6	5
		MEK5	7.4
		DDR1	12
		VEGFR2	18
		LOK	22
		SLK	60
KW-2449		DRAK1	2.9
		MLCK	4.3
		DRAK2	4.8
		YSK4	5.2
		BIKE	9.6
		MAP4K2	11
		LOK	13
		SLK	13
		FLT3	15
		SRPK2	15

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Lapatinib		EGFR	2.4
		ERBB2	7
		ERBB4	54
		PIK3C2B	670
		PIK4CB	940
		MEK5	1100
		SLK	3300
		RIPK2	3600
		LOK	4400
		MKK7	4400
LY-317615		GSK3B	8.3
		PRKCE	8.9
		PRKCD	25
		RSK4(Kin.Dom.1-N-terminal)	25
		PRKCQ	36
		PRKCH	46
		ERK8	76
		RSK2(Kin.Dom.1-N-terminal)	87
		DYRK1A	160
		PRKG2	170
LY-333531		PRKCQ	2.5
		PRKCD	3.6
		PRKCE	11
		PIM3	12
		YSK4	48
		HIPK3	77
		ERK8	88
		TAOK3	97
		HIPK2	140
		MAP3K3	170
MLN-120B		IKK-beta	19
		PIK4CB	270
		TYK2(JH2domain-pseudokinase)	500
		CLK1	640
		CLK4	1000
		IKK-alpha	1000
		DYRK1A	2100
		CLK2	2300
MLN-518		PDGFRA	2.4
		KIT	2.7
		FLT3	3
		PDGFRB	4.5
		CSF1R	4.9
		DDR2	120
		EGFR	410
		TRKA	450
		CLK1	630
		IRAK3	730
MLN-8054		AURKA	6.5
		DRAK2	8.1
		AURKC	26
		AURKB	43
		BLK	68
		DRAK1	190
		FGR	220
		YES	260
		TIE2	300
		EPHB1	330
Nilotinib		DDR1	1.1
		ABL1-nonphosphorylated	10
		ZAK	11
		ABL1-phosphorylated	13
		ABL2	26
		KIT	29
		DDR2	33
		p38-beta	36
		EPHA8	37
		CSF1R	45

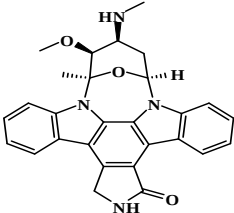
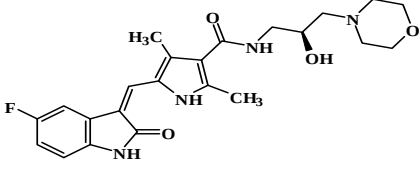
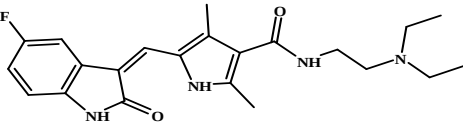
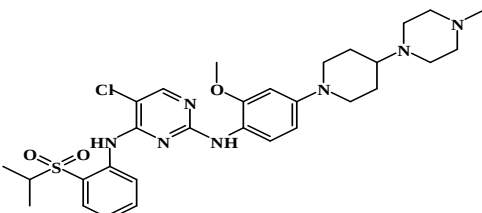
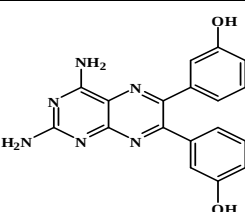
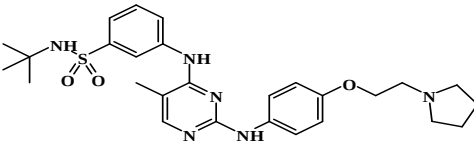
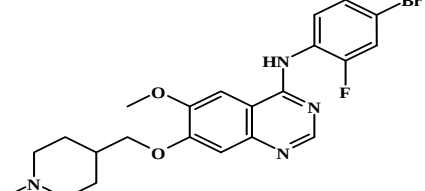
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Pazopanib		PDGFRB	2
		KIT	2.8
		PDGFRA	4.9
		CSF1R	7.9
		FLT1	14
		VEGFR2	14
		FLT4	27
		TAOK3	45
		DDR1	57
		EPHB6	81
PD-173955		ABL1-phosphorylated	0.58
		CSF1R	0.67
		ABL2	0.69
		SRC	0.71
		LCK	1.1
		PDGFRB	1.4
		ABL1-nonphosphorylated	1.5
		BLK	1.5
		KIT	1.8
		EPHB6	2
PHA-665752		MET	0.27
		SRPK1	41
		TAOK3	43
		BIKE	47
		GRK7	61
		HIPK3	63
		CAMKK1	64
		DDR1	70
		CAMKK2	73
		YSK4	78
PI-103		PIK3CA	1.5
		PIK3CB	1.7
		PIK3C2B	10
		MTOR	12
		PIK3CG	16
		PIK3CD	17
		PIK3C2G	41
		HIPK2	290
		HIPK3	310
		PIP5K2C	620
PKC-412		PKN1	9.3
		TBK1	9.3
		FLT3	11
		JAK3(JH1domain-catalytic)	12
		MLK1	15
		PKN2	15
		YSK4	15
		MLK3	17
		CAMK2A	20
		MARK3	21
PLX-4720		MEK5	0.16
		PFCDPK1(P.falciparum)	1.7
		SRMS	21
		ZAK	41
		BRK	48
		FGR	62
		RAF1	170
		KIT	180
		MEK4	190
		NEK11	190
PP-242		BMPR1B	2.1
		ACVRL1	2.9
		MTOR	3
		ACVR1	4
		RET	4.8
		YSK4	5.1
		MEK5	7.3
		ACVR2B	7.6
		PRKCE	9
		JAK2(JH1domain-catalytic)	11

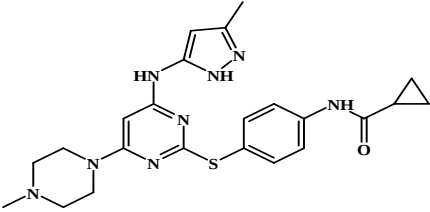
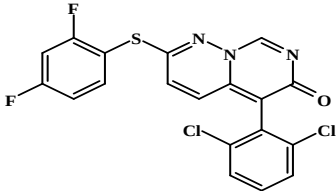
Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
PTK-787		KIT	5.1
		FLT1	9.6
		PDGFRB	25
		CSF1R	45
		VEGFR2	62
		PDGFRA	96
		DDR1	270
		FLT4	330
		CDK11	1500
		FRK	1800
R406		FLT3	0.71
		STK16	1.7
		GCN2(Kin.Dom.2,S808G)	3.3
		PDGFRB	3.3
		JAK2(JH1domain-catalytic)	3.5
		MLK2	3.8
		RET	4.1
		PLK4	4.2
		MLK1	4.3
		PLK3	5.1
R547		CDK2	0.53
		PCTK1	0.54
		CDK7	0.58
		CDK4-cyclinD1	0.61
		CDK4-cyclinD3	0.81
		PCTK2	0.86
		ICK	2.2
		CDK3	3.2
		PFTAIR2	7.2
		CDK5	7.4
SB-203580		p38-alpha	12
		GAK	19
		RIPK2	24
		NLK	25
		JNK3	35
		CSNK1D	37
		p38-beta	70
		CSNK1A1	75
		CSNK1E	100
		JNK2	130
SGX-523		MET	0.19
		DYRK1A	780
		DYRK1B	1800
		JNK3	1900
		PIP5K2C	3300
		JNK1	4200
		YSK4	6400
SKI-606		ABL1-phosphorylated	0.057
		ABL1-nonphosphorylated	0.12
		MAP4K5	0.5
		LCK	0.59
		ERBB3	0.77
		SRC	1
		GAK	1.3
		FRK	1.4
		ABL2	1.5
		STK35	2
Sorafenib		DDR1	1.5
		HIPK4	3.3
		ZAK	6.3
		DDR2	6.6
		FLT3	13
		RET	13
		CSF1R	28
		KIT	28
		FLT1	31
		PDGFRB	37

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
Staurosporine		SLK	0.024
		LOK	0.037
		CAMKK1	0.039
		SNARK	0.086
		PHKG2	0.14
		CAMK2A	0.16
		CAMKK2	0.16
		MST2	0.18
		MST1	0.19
		TAOK3	0.22
SU-14813		PDGFRB	0.29
		FLT3	0.54
		KIT	0.68
		PDGFRA	1.1
		VEGFR2	2.3
		CSF1R	3.6
		HUNK	3.7
		FLT1	4.7
		STK35	8.2
		YSK4	12
Sunitinib		PDGFRB	0.075
		KIT	0.37
		FLT3	0.41
		PDGFRA	0.79
		DRAK1	1
		VEGFR2	1.5
		FLT1	1.8
		CSF1R	2.5
		BIKE	5.5
		PHKG1	5.5
TAE-684		ROS1	0.49
		ULK1	0.83
		BMPR1B	0.85
		PLK4	0.93
		LTK	0.95
		ALK	1.1
		FAK	1.1
		PYK2	1.1
		SNARK	1.2
		FER	1.4
TG-100-115		PIK3C2G	3.2
		PIK3CG	5.3
		PIK3C2B	7.3
		TRPM6	7.9
		CLK2	43
		PIK3CA	59
		PIK3CB	80
		ADCK3	94
		RIPK4	97
		CLK4	130
TG-101348		GAK	1.1
		JAK2(JH1domain-catalytic)	1.1
		DAPK3	1.2
		STK16	6.6
		DCAMKL3	13
		FLT3	13
		DAPK1	16
		JAK1(JH1domain-catalytic)	18
		YSK4	19
		TYK2(JH1domain-catalytic)	21
Vandetanib		RIPK2	4.6
		EGFR	9.5
		DDR1	11
		ABL1-phosphorylated	16
		LCK	17
		RET	34
		ABL1-nonphosphorylated	48
		MEK5	49
		EPHA6	50
		STK35	56

Supplementary Table 2. Compound structures and highest affinity target kinases (excluding mutants)

Compound	Structure	Kinase	Kd, nM
VX-680/MK-0457		AURKA	3.9
		ABL2	4
		AURKC	6.3
		FLT3	6.5
		AURKB	7.4
		ABL1-phosphorylated	7.5
		PLK4	9.2
		ABL1-nonphosphorylated	13
		MLCK	15
		RIPK1	20
		p38-alpha	2.8
		p38-beta	74
VX-745		DDR1	1100
		FGR	1300
		YES	1600
		LYN	1700
		ABL2	1900
		FYN	2100
		CSF1R	2600
		BLK	3100